Министерство образования Республики Беларусь

Учреждение образования

«Брестский государственный технический университет»

Кафедра ИИТ

Лабораторная работа №4

За шестой семестр

По дисциплине: «Модели решения задач в ИС»

Тема: «Классификация линейно разделимых образов однослойным персептроном для логических функций n переменных»

Выполнил:
Студент 3 курса
Группы ИИ-26(1)
Пилипук М.А.
Проверил:
Якимук А.В.

Брест 2026

Цель: Реализовать нейронную сеть для решения задач синтеза логических функций от n входов. Сравнить эффективность обучения при различных начальных условиях.

N = 7, функция — инвертированное ИЛИ.

Угадайте где находится код программы.

Ради интереса и эксперимента будем использовать следующий «алгоритм» инверсии:

```
mask = np.array([0, 0, 0, 1, 1, 0, 0])
```

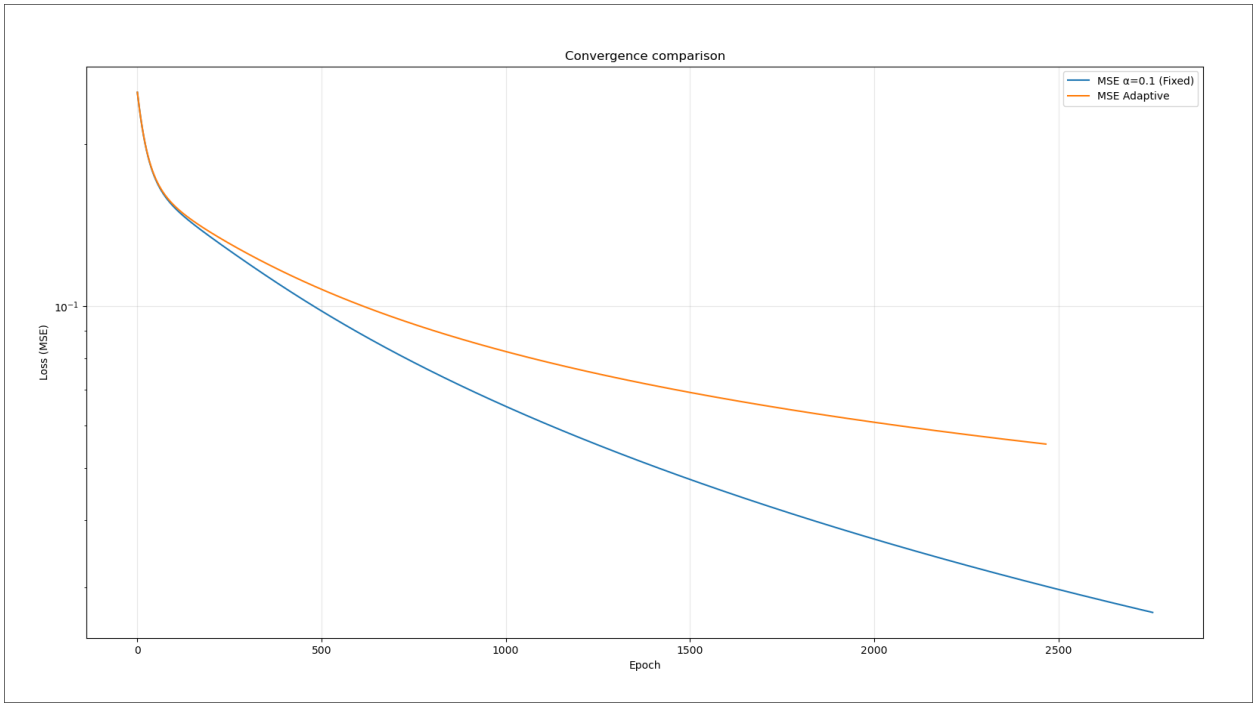

График зависимости MSE от эпохи.

MSE minimized at epoch = 2756
MSE minimized at epoch = 2466
Final Weights (Fixed): [-0.15245113 -0.15245113 -0.15245113 2.87416459 2.87416459 -0.15245113
 -0.15245113]
Final T (Fixed): 0.7261272898960764
Final Weights (Adapt): [-0.11190112 -0.11190112 -0.11190112 1.90613327 1.90613327 -0.11190112
 -0.11190112]
Final T (Adapt): 0.2778521771269554

Результаты проверки обобщающей способности сети:

Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.82320768]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.86331906]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.82320768]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.99113992]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98969583]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98969583]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98801921]%. Actually this is - [1.]

```
Ahmmf... Im thinking that this is 0 with [0.26288456]%. Actually this is - [0.]
Ahmmf... Im thinking that this is 0 with [0.23442694]%. Actually this is - [0.]
Ahmmf... Im thinking that this is 0 with [0.23442694]%. Actually this is - [0.]
Ahmmf... Im thinking that this is 0 with [0.20817987]%. Actually this is - [0.]
Ahmmf... Im thinking that this is 1 with [0.86331906]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.82320768]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.86331906]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.82320768]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.99113992]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98969583]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98969583]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98801921]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 0 with [0.23442694]%. Actually this is - [0.]
Ahmmf... Im thinking that this is 0 with [0.20817987]%. Actually this is - [0.]
Ahmmf... Im thinking that this is 0 with [0.20817987]%. Actually this is - [0.]
Ahmmf... Im thinking that this is 0 with [0.18416457]%. Actually this is - [0.]
Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.82320768]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.82320768]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.79991796]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.84431408]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.82320768]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.82320768]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.79991796]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98969583]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98801921]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98801921]%. Actually this is - [1.]
Ahmmf... Im thinking that this is 1 with [0.98607362]%. Actually this is - [1.]
```

Проглядывается интересная закономерность, что единицы, идущие прямо после нулей, имеют меньший шанс уверенности, чем единицы, находящиеся перед нулями. Учитывая что паттерн в целом повторяется, а количество 0 — довольно низкое, то очевидно, что будет подобный результат.

Вывод: реализовал нейронную сеть для синтеза логической функции от n входов.